

LECTURE

File Structure and Organization



Organizing Code



customer-app.tf

Infrastructure Requirements:

- ☐ S3 Bucket
- ☐ Virtual Machine
- ☐ Subnet
- ☐ Firewall Configuration
- ☐ DNS Record
- ☐ Load Balancer
- ☐ Kubernetes Cluster

Terraform Configuration File

variable

variable

resource

resource

resource

resource

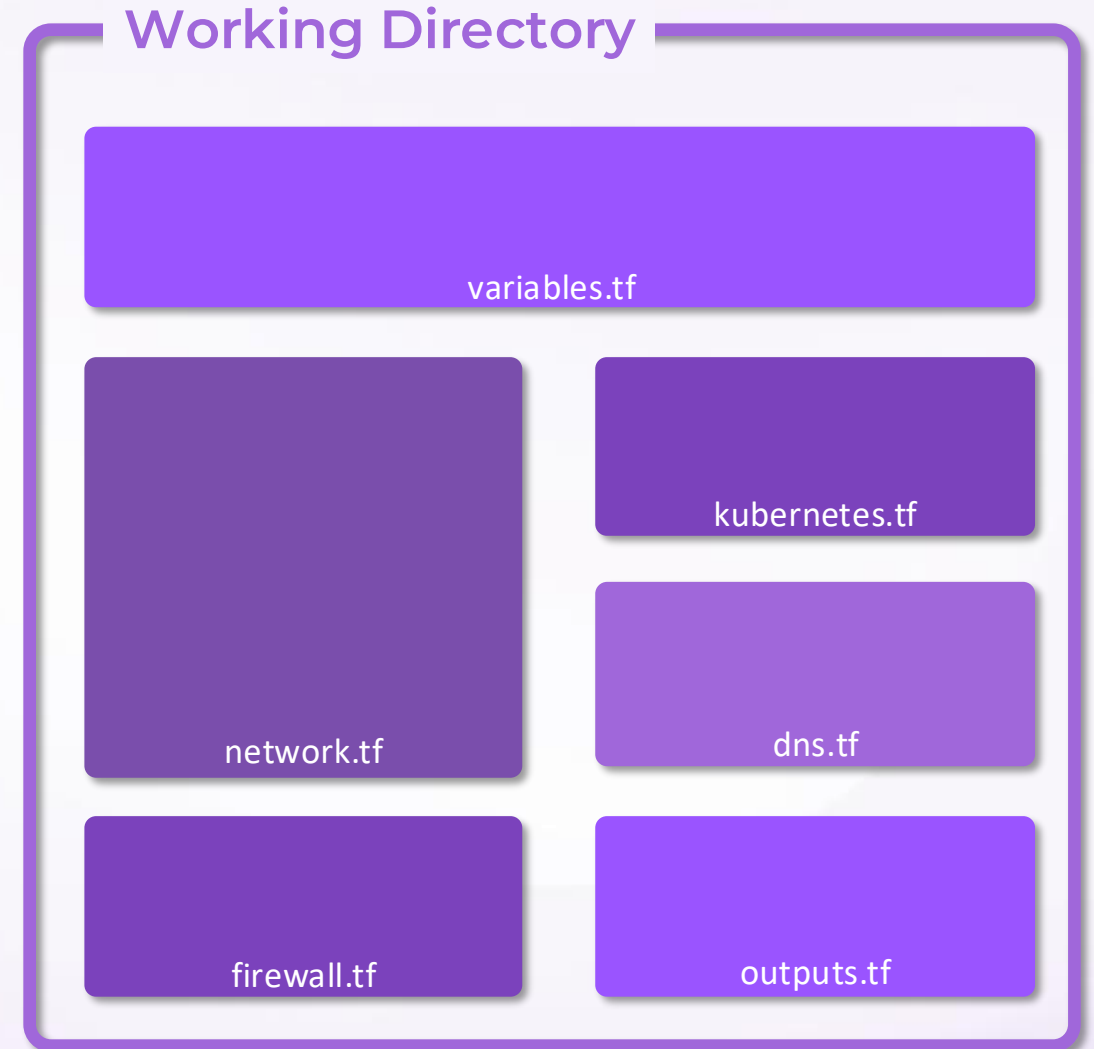
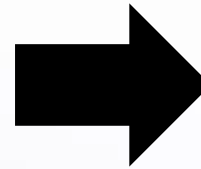
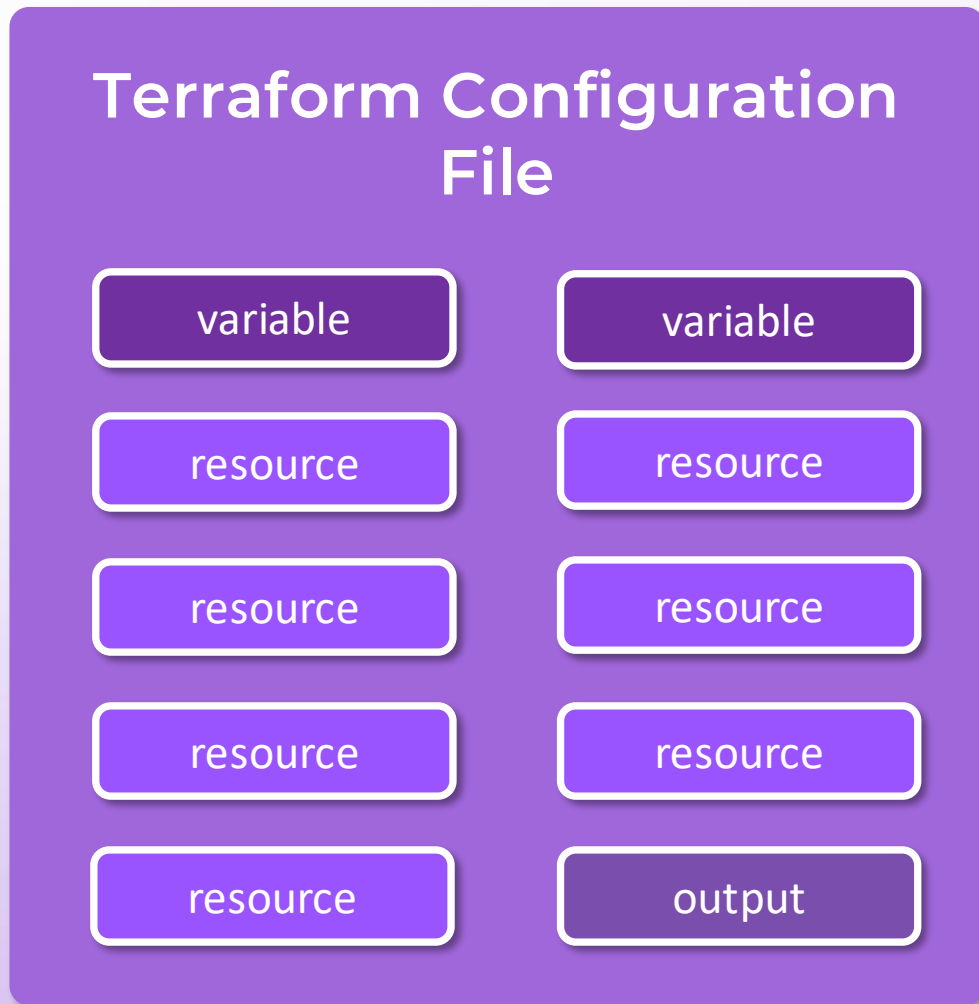
resource

resource

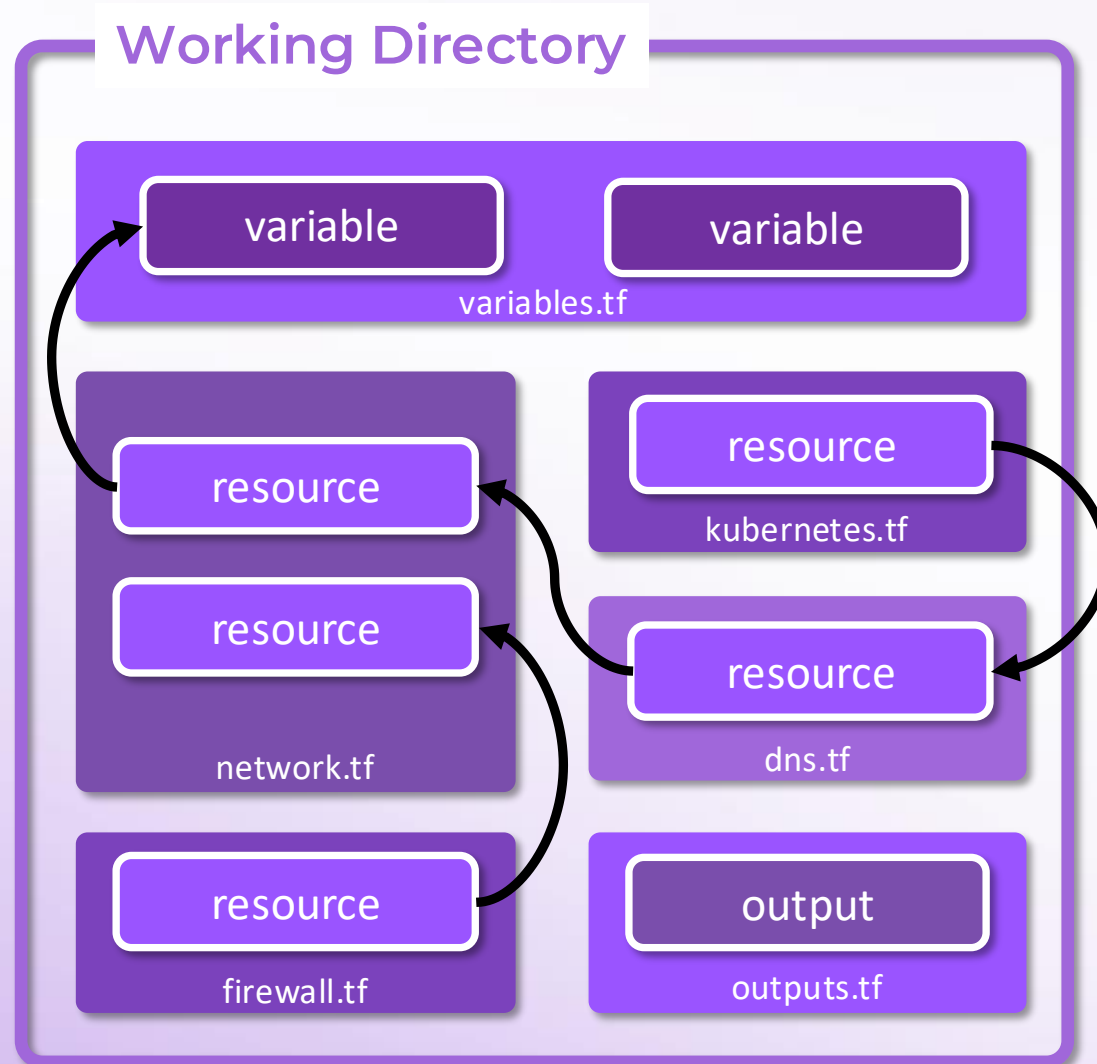
resource

output

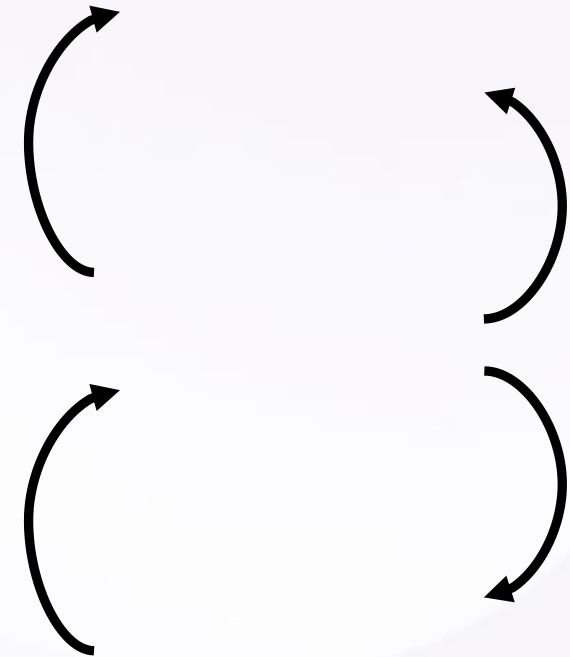
Organizing Code



Executing Code



Terraform Processes All Files Together



Common Terraform Files



main.tf

Commonly used for primary infrastructure components



variables.tf

Variable definitions used throughout your infrastructure



outputs.tf

Output blocks to output values to the terminal or to use for inputs to other modules



providers.tf

Provider configurations and requirements



terraform.tfvars

Define the values for your variables – usually ignored by .gitignore



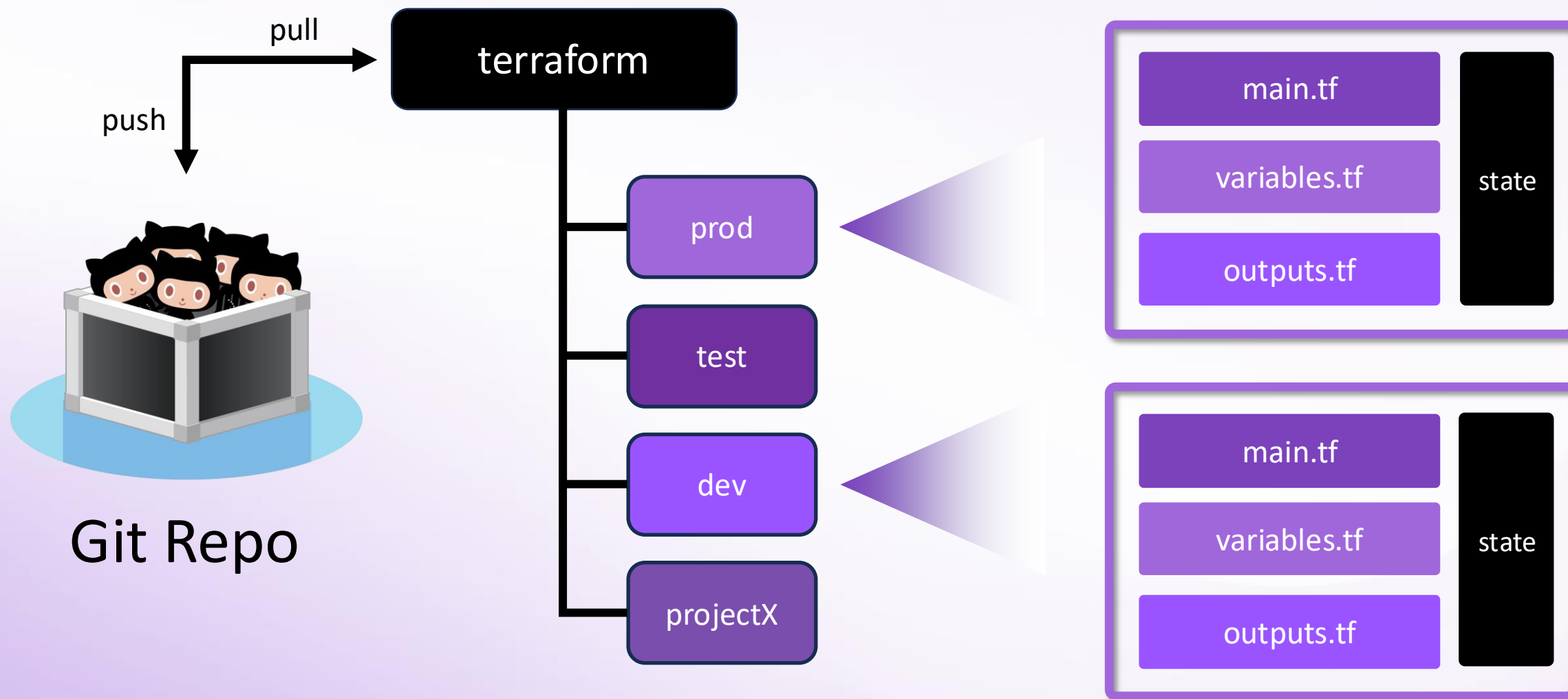


Additional Files

- `terraform.tfstate`
The file that Terraform uses to store state
- `terraform.tfstate.backup`
Backup of the previous state file before Terraform writes anything new to the state file
- `.terraform.lock.hcl`
Used to track and select provider versions
- `.gitignore`
Instructs Git to ignore files included in this file

```
▼ aws_deploy
  > .terraform
  > scripts
  [H] .terraform.lock.hcl
  [Y] main.tf
  [Y] outputs.tf
  [Y] providers.tf
  [Y] terraform.tfstate
  [Y] terraform.tfstate.backup
  [Y] variables.tf
```

Working Directory



SUMMARY



Terraform only processes the files in the current working directory unless you reference other files



While not required, there are common file names for Terraform configurations



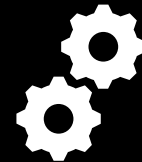
You can organize your files across multiple .tf files



Terraform creates and manages files for state and other configurations



Terraform combines all the files in the working directory when executing



Use subdirectories to break up your Terraform to better organize and reduce the blast radius of changes